Reinforcement Learning

End Semester Exam

04/12/2025

Sanjit K. Kaul

Instructions: You have about 120 minutes to work on the questions. Answers with no supporting steps will receive zero credit. No resources, other than a pen/pencil, are allowed. In case you believe that required information is unavailable, make a suitable assumption.

Question 1. 25 **marks** The policy gradient theorem for stochastic policies states that the gradient of the performance measure $J(\theta)$ with respect to the parameter vector θ is

$$\nabla_\theta J(\theta) \propto \sum_s \mu(s) \sum_a \nabla_\theta \pi_\theta(a|s) q_{\pi_\theta}(s,a).$$

Consider

$$\widehat{\nabla}_\theta J(\theta) = \nabla_\theta \log(\pi_\theta(A_t|S_t))\, A_{\pi_\theta}(S_t, A_t).$$

Show that $\widehat{\nabla}_\theta J(\theta)$ is a sample estimate of the gradient. Specifically, show that the expected value of $\widehat{\nabla}_\theta J(\theta)$ is equal to $\nabla_\theta J(\theta)$.

In practice, the advantage $A_{\pi_\theta}(S_t, A_t)$ must be estimated using a value network parametrized by vector w. Calculate a suitable n-step, where $n \geq 1$, sample estimate of the advantage. Your steps must clearly show that what you have calculated is in fact a sample estimate. Use the sample estimate to calculate the loss function of the value network (parameterized by vector w) and use the loss function to update w.

Question 2. 25 **marks** Calculate the gradient, with respect to the parameter vector θ, of the performance measure

$$J_b(\pi_\theta) = E_{S\sim\rho^b, A\sim\pi_\theta}\left[Q_{\pi_\theta}(S, A)\right],$$

where b is a behavior policy independent of θ. The unnormalized state distribution ρ^b is defined for every state s as $\rho^b(s) = \sum_{t=0}^{\infty} \gamma^t P[S_t = s]$. In your calculations, assume that small enough (infinitesimal) changes in θ do not change the action-values. The final expression for the gradient must be written as an expected value with the policy and state distributions clearly specified. Provide an update of θ using a sample estimate of the expected value. Note that episodes are generated using the behavior policy b.

Question 3. 15 **marks** Consider the MDP in Figure 1. Propose a linear architecture, with a small enough parameter vector, that computes the optimal values of all non-terminal states in the MDP.

Question 4. 15 **marks** Consider the MDP in Figure 2. You use exact policy iteration to calculate an optimal policy with the goal of following the shortest path (smallest number of arcs) from any starting

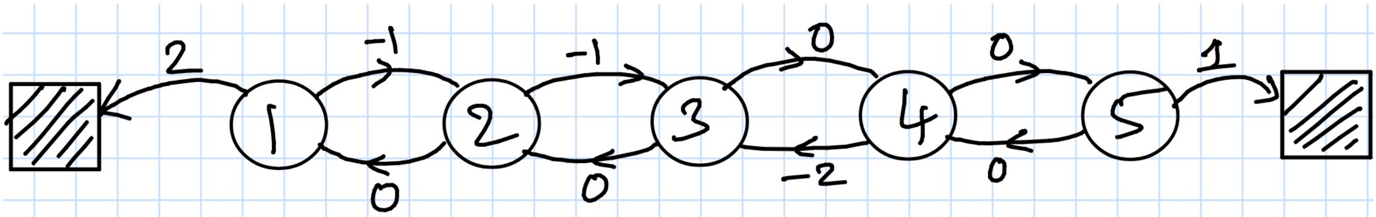

Fig. 1: The arrows show all possible state transitions. The numbers on the arrows are the corresponding rewards.

state to the terminal state. You set $\gamma = 1$ and start with an initial policy that chooses all outgoing arcs from a state with equal probability. Policy iteration ends with a policy that in state k, $k = 0, 1, 2, 3, 4$, chooses an arc that transitions the state to $k + 1$. In state 5, it chooses the arc that takes us to the terminal state. While the policy results in the maximum value of 1 for states $0, \ldots, 5$, it doesn't pick the shortest number of arcs to the terminal state in state 4. How will you fix this problem? Show that your proposal will ensure shortest number of arcs to the terminal state from any starting state.

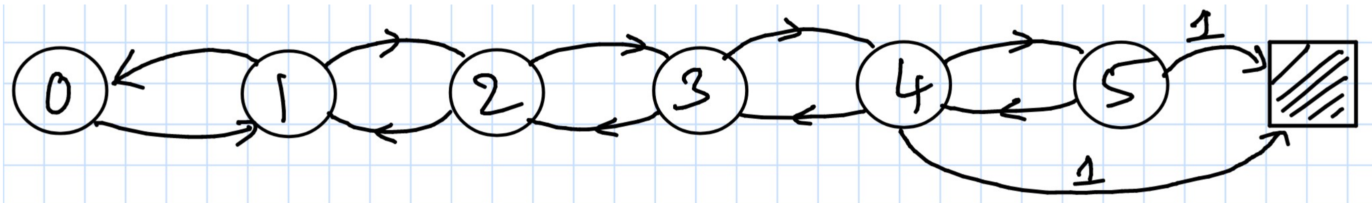

Fig. 2: The arrows show all possible state transitions. All transitions other than the one that goes into the terminal state result in a reward of 0. The transition that ends in the terminal state gives a reward of 1.

Question 5. 10 **marks** The λ-return is defined as

$$G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_{t:t+n},$$

where $G_{t:t+n}$ is the n-step return. Show that G_t^1, which is the λ-return obtained in the limit as $\lambda \to 1$, is equal to the Monte Carlo return G_t.

Question 1. 25 **marks** The policy gradient theorem for stochastic policies states that the gradient of the performance measure $J(\theta)$ with respect to the parameter vector θ is

$$\nabla_\theta J(\theta) \propto \sum_s \mu(s) \sum_a \nabla_\theta \pi_\theta(a|s) q_{\pi_\theta}(s,a).$$

Consider

$$\widehat{\nabla}_\theta J(\theta) = \nabla_\theta \log(\pi_\theta(A_t|S_t))\, A_{\pi_\theta}(S_t, A_t).$$

Show that $\widehat{\nabla}_\theta J(\theta)$ is a sample estimate of the gradient. Specifically, show that the expected value of $\widehat{\nabla}_\theta J(\theta)$ is equal to $\nabla_\theta J(\theta)$. ← (a) 9/25

(b) 8/25

In practice, the advantage $A_{\pi_\theta}(S_t, A_t)$ must be estimated using a value network parametrized by vector w. Calculate a suitable n-step, where $n \geq 1$, sample estimate of the advantage. Your steps must clearly show that what you have calculated is in fact a sample estimate. Use the sample estimate to calculate the loss function of the value network (parameterized by vector w) and use the loss function to update w. (c) 8/25

(a) The expected value of $\hat{\nabla}_\theta J(\theta)$ is

$$E[\hat{\nabla}_\theta J(\theta)] = E[\nabla_\theta \log \pi_\theta(A_t|S_t)\; A_{\pi_\theta}(S_t,A_t)]$$

$$= \sum_s \sum_a \nabla_\theta \log \pi_\theta(a|s)\; A_{\pi_\theta}(s,a)\; P_{\pi_\theta}[S_t=s, A_t=a]$$

$$= \sum_s \sum_a \nabla_\theta \log \pi_\theta(a|s)\; A_{\pi_\theta}(s,a)\; P_{\pi_\theta}[S_t=s]\; \pi_\theta(a|s)$$

$$= \sum_s \sum_a \nabla_\theta \log \pi_\theta(a|s)\; (q_{\pi_\theta}(s,a) - v_{\pi_\theta}(s))\; P_{\pi_\theta}[S_t=s]\; \pi_\theta(a|s)$$

$$= \sum_s \sum_a \nabla_\theta \log \pi_\theta(a|s)\; q_{\pi_\theta}(s,a)\; P_{\pi_\theta}[S_t=s]\; \pi_\theta(a|s)$$
$$- \sum_s \sum_a v_{\pi_\theta}(s)\; \nabla_\theta \log \pi_\theta(a|s)\; P_{\pi_\theta}[S_t=s]\; \pi_\theta(a|s)$$

$$= \sum_s \sum_a \nabla_\theta \log \pi_\theta(a|s)\; q_{\pi_\theta}(s,a)\; P_{\pi_\theta}[S_t=s]\; \pi_\theta(a|s)$$
$$- \sum_s v_{\pi_\theta}(s) \left[\sum_a \nabla_\theta \pi_\theta(a|s)\right] P_{\pi_\theta}[S_t=s]$$

Expanding the expected value to end up with a sum over $q()$ & a sum over $v()$. Up to 5/9

$$= \sum_s \sum_a \nabla_\theta \log \pi_\theta(a|s)\; q_{\pi_\theta}(s,a)\; P_{\pi_\theta}[S_t=s]\; \pi_\theta(a|s)$$

$$\left[\text{Since } \sum_a \nabla_\theta \pi_\theta(a|s) = \nabla_\theta \sum_a \pi_\theta(a|s) = \nabla_\theta(1) = 0.\right]$$

$$= E[\nabla_\theta \log \pi_\theta(A_t|S_t)\; q_{\pi_\theta}(S_t,A_t)].$$

Make the observation about how the $v_{\pi_\theta}(s)$ term is zeroed. Up to 4/9

$\therefore\ \hat{\nabla}_\theta J(\theta)$ is a sample estimate of $\nabla_\theta J(\theta)$. Specifically,

$$E[\hat{\nabla}_\theta J(\theta)] = \underline{\nabla_\theta J(\theta)}.$$

(b) For any (s,a)

$$A_{\pi_\theta}(s,a) = q_{\pi_\theta}(s,a) - v_\pi(s)$$

$$= E_{\pi_\theta}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t=s, A_t=a] - v_\pi(s).$$

$$= E_{\pi_\theta}[R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n v_\pi(S_{t+n}) \mid S_t=s, A_t=a]$$
$$- v_\pi(s).$$

Wrote the advantage fn in terms of the n-step return. Up to 6/8

A sample estimate of the advantage $A_{\pi_\theta}(s,a)$ is

$$\hat{A}_{\pi_\theta}(s,a) = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n v_\pi(S_{t+n}) - v_\pi(s),$$

where this estimate is calculated using the observed sequence of rewards and the state S_{t+n}.

Given the above part is correctly shown, up to 2/8. If above is incorrect, this part gets 0.

(c) Loss function of the value network:

The n-step target is $R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n v_w(S_{t+n})$

The current estimate is $v_w(S_t)$. Here w parametrizes the value network.

Loss is

$$\left(R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n v_w(S_{t+n}) - v_w(S_t)\right)^2$$

Clearly stating the loss. Up to 4/8

The gradient (specifically, semi-gradient) w.r.t. w is:

$$-2\left(R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n v_w(S_{t+n}) - v_w(S_t)\right) \nabla v_w(S_t)$$

Calculating the semi-gradient 2/8

$\boxed{\text{Update of } w\text{: (gradient descent)}}$

$$w = w - \alpha\left(R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n v_w(S_{t+n}) - v_w(S_t)\right) \nabla v_w(S_t)$$

Correct gradient descent step. 2/8

Question 2. 25 **marks** Calculate the gradient, with respect to the parameter vector θ, of the performance measure

$$J_b(\pi_\theta) = E_{S\sim\rho^b, A\sim\pi_\theta}\left[Q_{\pi_\theta}(S,A)\right],$$

where b is a behavior policy independent of θ. The unnormalized state distribution ρ^b is defined for every state s as $\rho^b(s) = \sum_{t=0}^{\infty}\gamma^t P[S_t = s]$. In your calculations, assume that small enough (infinitesimal) changes in θ do not change the action-values. The final expression for the gradient must be written as an expected value with the policy and state distributions clearly specified. Provide an update of θ using a sample estimate of the expected value. Note that episodes are generated using the behavior policy b.

$$J_b(\pi_\theta) = E_{S\sim\rho^b,\, A\sim\pi_\theta}\left[Q_{\pi_\theta}(S,A)\right]$$

$$= \sum_s \sum_a q_{\pi_\theta}(s,a)\, \pi_\theta(a|s)\, \rho^b(s)$$

$$\nabla_\theta J_b(\pi_\theta) = \sum_s \sum_a \rho^b(s)\, \nabla_\theta\left[q_{\pi_\theta}(s,a)\, \pi_\theta(a|s)\right]$$

$$= \sum_s \sum_a \rho^b(s) \left[\pi_\theta(a|s)\, \nabla_\theta q_{\pi_\theta}(s,a) + q_{\pi_\theta}(s,a)\, \nabla_\theta \pi_\theta(a|s)\right]$$

Correct expansion of the gradient. The uv-rule correctly applied 5/25

$$\approx \sum_s \sum_a \rho^b(s)\, q_{\pi_\theta}(s,a)\, \nabla_\theta \pi_\theta(a|s)$$

(We approximate as per assumption)

Approximation 5/25

$$= \sum_s \sum_a \rho^b(s)\, q_{\pi_\theta}(s,a)\, \nabla_\theta \pi_\theta(a|s)\, \frac{b(a|s)}{b(a|s)}$$

$$= \sum_s \sum_a \frac{1}{b(a|s)}\, q_{\pi_\theta}(s,a)\, \nabla_\theta \pi_\theta(a|s)\, \rho^b(s)\, b(a|s)$$

$$= E_b\left[\frac{1}{b(a|s)}\, q_{\pi_\theta}(s,a)\, \nabla_\theta \pi_\theta(a|s)\right]$$

$$= E_b\left[\frac{\pi_\theta(a|s)}{b(a|s)}\, q_{\pi_\theta}(s,a)\, \nabla_\theta \ln \pi_\theta(a|s)\right]$$

They are exactly the same. Either form or both is fine.

Since data/episodes are generated using the behavior policy $b(a|s)$, we must rewrite the double sum as an expectation using policy b.

Deriving the expectation using policy $b()$. Up to 10/25

A sample estimate $\nabla_\theta J_b(\pi_\theta)$ is therefore:

$$\hat{\nabla}_\theta J_b(\pi_\theta) = \frac{\pi_\theta(a|s)}{b(a|s)}\, q_{\pi_\theta}(s,a)\, \nabla_\theta \ln \pi_\theta(a|s).$$

Sample estimate Up to 2.5/25

We want to maximize $J_b(\pi_\theta)$. Gradient ascent step is

$$\theta = \theta + \alpha\, \hat{\nabla}_\theta J_b(\pi_\theta)$$

$$= \theta + \alpha\, \frac{\pi_\theta(a|s)}{b(a|s)}\, q_{\pi_\theta}(s,a)\, \nabla_\theta \ln \pi_\theta(a|s).$$

Gradient ascent step 2.5/25

Question 3. 15 **marks** Consider the MDP in Figure 1. Propose a linear architecture, with a small enough parameter vector, that computes the optimal values of all non-terminal states in the MDP.

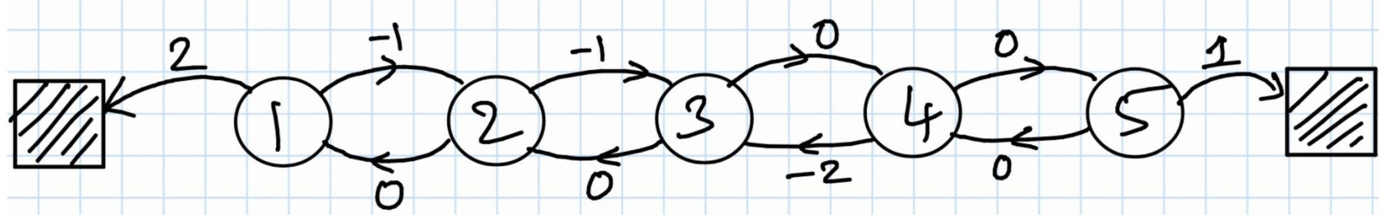

Fig. 1: The arrows show all possible state transitions. The numbers on the arrows are the corresponding rewards.

The optimal values are easy to decipher for the MDP. Here we won't worry about showing these are optimal.

$V_*(1)=2$

$V_*(2)=0+V_*(1)=2$ [The only +ve reward going right is from 5 to terminal & it is 1]

$V_*(3)=0+V_*(2)=2$

$V_*(4)=0+V_*(5)=0+1=1.$

$V_*(5)=1.$

Calculating the optimal values. Up to 5/15

The value function is 5×1.

A linear architecture requires a (feature) matrix Φ and a weight vector w.

We require
$$\Phi w = \begin{bmatrix} V_*(1) \\ V_*(2) \\ V_*(3) \\ V_*(4) \\ V_*(5) \end{bmatrix}_{5\times 1} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}$$

Note that $V_*(1)=V_*(2)=V_*(3)=2$ and $V_*(4)=V_*(5)=1.$

Set
$$\Phi = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}_{5\times 2}.$$

Observe that $w=\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{2\times 1}$ gives $\Phi w = \begin{bmatrix} V_* \end{bmatrix}.$

Given the calculated optimal values (even if wrong), reasoning of the provided kind to arrive at the feature vector matrix Φ (Up to 5/15) & the weight vector w (Up to 5/15).

Question 4. 15 **marks** Consider the MDP in Figure 2. You use exact policy iteration to calculate an optimal policy with the goal of following the shortest path (smallest number of arcs) from any starting state to the terminal state. You set $\gamma = 1$ and start with an initial policy that chooses all outgoing arcs from a state with equal probability. Policy iteration ends with a policy that in state k, $k = 0, 1, 2, 3, 4$, chooses an arc that transitions the state to $k + 1$. In state 5, it chooses the arc that takes us to the terminal state. While the policy results in the maximum value of 1 for states $0, \dots, 5$, it doesn't pick the shortest number of arcs to the terminal state in state 4. How will you fix this problem? Show that your proposal will ensure shortest number of arcs to the terminal state from any starting state.

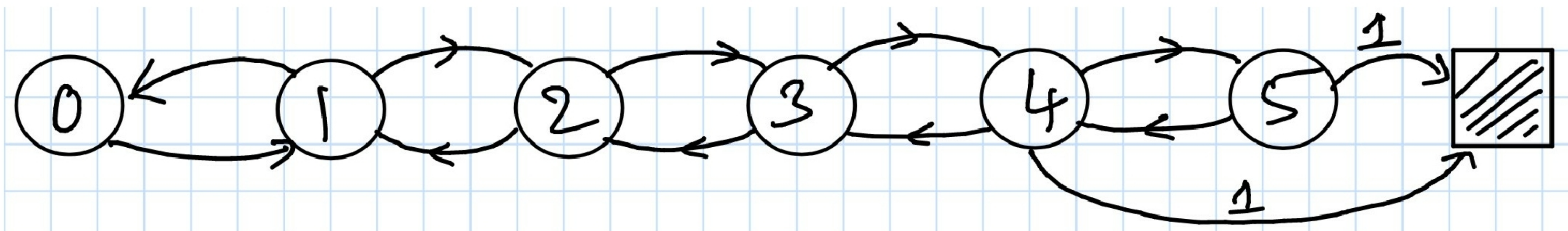

Fig. 2: The arrows show all possible state transitions. All transitions other than the one that goes into the terminal state result in a reward of 0. The transition that ends in the terminal state gives a reward of 1.

You can't change the MDP.

Consider the return of the path (sequence of edges/arrows) $2 \to 3 \to 4 \to 5 \to$ ▨. The return, given $\gamma = 1$, is 1.

Now consider the return of the path $2 \to 3 \to 4 \to$ ▨. The return is still 1. However, the number of arcs is 3 instead of the 4

To rectify the problem, a reward of 1 that appears later in time should contribute less than 1 to the return. Later in time, as in after a larger number of arcs.

Therefore, setting $0 < \gamma < 1$ solves the problem.

Any $\gamma \in (0,1)$ is a good choice to achieve what we want.

Let $\gamma = 0.99$.

$2 \to 3 \to 4 \to 5 \to$ ▨, gives a return of $0 + 0 + 0 + \gamma^3(1) = (0.99)^3$.

$2 \to 3 \to 4 \to$ ▨, gives $0 + 0 + \gamma^2(1) = (0.99)^2 > (0.99)^3$.

It is easy to show this for every starting state.

Reasoning that identifies (via examples) or by any other means that γ must be changed to some value in $(0,1)$. Up to 10/15.

Demonstration that the chosen γ fixes the issue at hand.

5/15

Question 5. 10 **marks** The λ-return is defined as

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1}G_{t:t+n},$$

where $G_{t:t+n}$ is the n-step return. Show that G_t^1, which is the λ-return obtained in the limit as $\lambda \to 1$, is equal to the Monte Carlo return G_t.

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1}R_{t+n} + \gamma^n V(S_{t+n})$$

$$G_t^\lambda = (1-\lambda)\left[G_{t:t+1} + \lambda G_{t:t+2} + \lambda^2 G_{t:t+3} + \cdots\right]$$

$$= (1-\lambda)\Big[R_{t+1} + \gamma V(S_{t+1})$$
$$+ \lambda\left(R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})\right)$$
$$+ \lambda^2\left(R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 V(S_{t+3})\right)$$
$$\vdots\Big]$$

$$= (1-\lambda)\Big[(1+\lambda+\lambda^2+\cdots)R_{t+1} + \gamma\lambda(1+\lambda+\lambda^2+\cdots)R_{t+2}$$
$$+ \gamma^2\lambda^2(1+\lambda+\lambda^2+\cdots)R_{t+3} + \cdots\Big]$$

$$+ (1-\lambda)\gamma\left[V(S_{t+1}) + \lambda\gamma V(S_{t+2}) + \lambda^2\gamma^2 V(S_{t+3}) + \cdots\right]$$

$$= R_{t+1} + \gamma\lambda R_{t+2} + \gamma^2\lambda^2 R_{t+3} + \cdots$$
$$+ (1-\lambda)\gamma\left[V(S_{t+1}) + \lambda\gamma V(S_{t+2}) + \lambda^2\gamma^2 V(S_{t+3}) + \cdots\right]$$

A clean and correct expansion of G_t^λ: Up to 8/10

In the limit as $\lambda \to 1$, for $\gamma < 1$, we get

$$\lim_{\lambda\to 1} G_t^\lambda = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$$
$$= G_t. \text{ (The Monte Carlo return)}$$

Taking the limit to arrive at the final result 2/10

Question 6. 10 **marks** Calculate the optimal value function for the MDP in Figure 1.

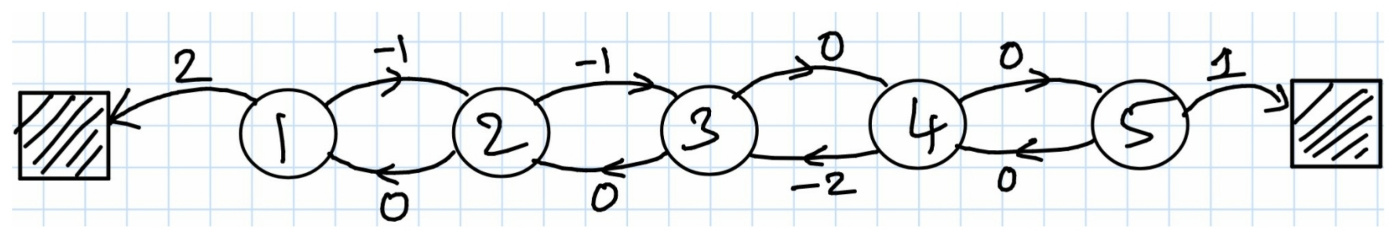

Fig. 1: The arrows show all possible state transitions. The numbers on the arrows are the corresponding rewards.

We start with a guess:

$v(1)=2$
$v(2)=2$
$v(3)=2$
$v(4)=1$
$v(5)=1.$

We must show that our guess is the optimal value function.

Guess $2/10$

The optimal value function must satisfy the Bellman Optimality equations.

Sufficient condition $2/10$

The Bellman Optimality equations are:

$$v_*(1) = \max(2, -1+v_*(2))$$
$$v_*(2) = \max(v_*(1), -1+v_*(3))$$
$$v_*(3) = \max(v_*(2), v_*(4))$$
$$v_*(4) = \max(-2+v_*(3), v_*(5))$$
$$v_*(5) = \max(v_*(4), 1).$$

Stating the optimality equations $5/10$

Note that $v(\,)$ satisfies these equations.

We have:

$$v(1) = \max(2, -1+v(2)) = \max(2, -1+2) = 2.$$

$v(2) = \max(2, -1+2) = 2$ (our guess)

$v(3) = \max(2, 1) = 2 = v(3) \leftarrow$ our guess

$v(4) = \max(-2+2, 1) = 1 = v(4)$

$v(5) = \max(1, 1) = 1 = v(5)$ (our guess)

Ensuring the guess is in fact optimal $1/10$

Any other valid method that results in the optimal function, like policy iteration, is fine.

Only a correct guess gives $2/10$ %.